List 12. Multinomial Logit

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Contents

1 Regression fitting

#**1**. For the dataset `Diamond` consider a regression

`certification on price & carat`.

1. write down the specification
2. Evaluate matrices of regression design $\boldsymbol{y}, \boldsymbol{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)

4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

#**2**. For the dataset `Diamond` consider a regression

`colour on log(price) & carat`.

1. write down the specification
2. Evaluate matrices of regression design $\boldsymbol{y}, \boldsymbol{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)
4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

#**3**. For the dataset `Diamond` consider a regression

`clarity on price, carat, carat`^{2}.

1. write down the specification
2. Evaluate matrices of regression design $\boldsymbol{y}, \boldsymbol{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)
4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

#**4**. For the dataset `diamonds` consider a regression

`cut on log(price), carat, x, y, z`.

1. write down the specification

2. Evaluate matrices of regression design $\boldsymbol{y}, \boldsymbol{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)
4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

#**5**. For the dataset `diamonds` consider a regression

`color on log(price), carat, carat`^{2}`, x, y, z`.

1. write down the specification
2. Evaluate matrices of regression design $\boldsymbol{y}, \boldsymbol{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)
4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

#**6**. For the dataset `diamonds` consider a regression

`clarity on log(price), log`^{2}`(price), carat, x, y, z`.

1. write down the specification
2. Evaluate matrices of regression design $\boldsymbol{y}, \boldsymbol{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)
4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

2 z-test

For each text calculate related critical values.

2.1 Significance

#**1**. For the dataset `Diamond` consider a regression

`certification on price & carat`.

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#**2**. For the dataset `Diamond` consider a regression

`colour on price & carat`.

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#**3**. For the dataset `Diamond` consider a regression

`clarity on price & carat`.

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance

3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#**4**. For the dataset `diamonds` consider a regression

`cut on log(price), carat, x, y, z`.

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#**5**. For the dataset `diamonds` consider a regression

`color on log(price), carat, x, y, z`.

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#**6**. For the dataset `diamonds` consider a regression

`clarity on log(price), carat, x, y, z`.

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

3 LR/Wald tests

For each text calculate related critical values.

3.1 Overall significance

#**1**. For the dataset `Diamond` consider a regression

`certification on price & carat`.

Test overall significance of the regression and state the testing hypothesis

#**2**. For the dataset `Diamond` consider a regression

`colour on price & carat`.

Test overall significance of the regression and state the testing hypothesis

#**3**. For the dataset `Diamond` consider a regression

`clarity on price & carat`.

Test overall significance of the regression and state the testing hypothesis

#**4**. For the dataset `diamonds` consider a regression

`cut on log(price), carat, x, y, z`.

Test overall significance of the regression and state the testing hypothesis

#**5**. For the dataset `diamonds` consider a regression

`color on log(price), carat,` $\texttt{carat}^2$`, x, y, z`.

Test overall significance of the regression and state the testing hypothesis

#**6**. For the dataset `diamonds` consider a regression

`clarity on log(price),` $\texttt{log}^2$`(price), carat, x, y, z`.

Test overall significance of the regression and state the testing hypothesis

3.2 Joint significance

#**7**. For the dataset `diamonds` consider a regression

$$\texttt{cut on log(price), carat, x, y, z}.$$

Test joint significance of regressors `x, y, z` and state the testing hypothesis

#**8**. For the dataset `sleep75` consider a regression

$$\texttt{cut on log(price), carat, carat}^2\texttt{, x, y, z}.$$

Test the significance of `carat` and state the testing hypothesis

#**9**. For the dataset `sleep75` consider a regression

$$\texttt{cut on log(price), log}^2\texttt{(price), carat, x, y, z}.$$

Test the significance of `carat` and state the testing hypothesis

4 Confidence intervals

#**1**. For the dataset `Diamond` consider a regression

$$\texttt{certification on price \& carat}.$$

Evaluate 90%-level confidence intervals for coefficients.
Which are significant at 10%-significant level?

#**2**. For the dataset `diamonds` consider a regression

$$\texttt{cut on log(price), carat, x, y, z}.$$

Evaluate 95%-level confidence intervals for coefficients.
Which are significant at 5%-significant level?

#**3**. For the dataset `diamonds` consider a regression

$$\texttt{color on log(price), carat, x, y, z, x}^2\texttt{, y}^2\texttt{, z}^2.$$

Evaluate 99%-level confidence intervals for coefficients.
Which are significant at 1%-significant level?

5 Prediction

#1. For the dataset `Diamond` consider a regression

`certification on price & carat`.

Consider diamonds with characteristics

	price	carat
1	1260	0.3
2	1427	0.32
3	5193	0.71
4	9573	1.01

#2. For the dataset `Diamond` consider a regression

`colour on log(price), carat,` $\texttt{carat}^2$.

Consider diamonds with characteristics

	price	carat
1	1555	0.31
2	6709	0.89
3	5030	0.73
4	9302	1.06

#3. For the dataset `Diamond` consider a regression

`clarity on log(price),` $\texttt{log}^2$`(price), carat`.

Consider diamonds with characteristics

	price	carat
1	1126	0.31
2	1911	0.4
3	3651	0.57
4	9153	1.01